

Module-wise Question Paper Analysis - NLP (DLOC-III)

Module	Question	Month & Year	Marks	Frequency
Module 1: Introduction to NLP	Explain the different stages involved in NLP process with example	Dec 2022	10	2
	Discuss the challenges in various stages of NLP	Dec 2023	10	1
	Explain the challenges of Natural Language Processing	May 2023	5	1
	Explain the various stages of NLP	May 2023	5	1
	Explain the applications of NLP	May 2024	5	1
	Explain ambiguities associated at each level with example	May 2023	10	1
Module 2: Word Level Analysis	Define affixes. Explain the types	Dec 2022	5	1
	Morphological analysis & FST role	May 2023	10	1
	Inflectional and derivational morphology with example	May 2024	10	1
	Porter stemming algorithm with rules / in detail	Multiple (2022, 2023, 2024)	10	4
	Explain perplexity of a language model	Dec 2022	5	1

	N-gram model in spelling correction	May 2023	5	1
	Bigrams and predict next word (corpus-based)	Dec 2022	5	1
	HMM: transition/emission + POS tagging	May 2023	10	1
	Explain Good Turing Discounting	Dec 2023	5	1
Module 3: Syntax Analysis	What is POS tagging? Challenges?	Dec 2022	10	2
	Explain challenges in POS tagging	Dec 2023	10	2
	HMM for POS tagging	May 2024	10	1
	Max Entropy model for POS tagging	May 2023	10	1
	CRF for sequence labeling	Dec 2023, May 2024	10	2
	Use of PCFG with example	Dec 2023	10	1
	CYK parsing for sentence	May 2023	10	1
	Compare top-down vs bottom-up parsing	Dec 2022	10	1
Module 4: Semantic Analysis	WSD: dictionary-based approach	Dec 2022, May 2023	10	2
	Lesk algorithm for WSD with example	Dec 2023, May 2024	10	2
	Lexical semantic analysis with example	May 2024	10	1
	Semantic analysis in NLP	May 2024	5	1
Module 5:	Hobbs	Dec 2022	10	1

Pragmatic & Discourse Processing	algorithm for pronoun resolution			
	Centering algorithm for reference resolution	May 2023	10	1
	Reference resolution	Dec 2023	5	1
	Types of referents complicating resolution	May 2023	5	1
	Five types of referring expressions	Dec 2023	10	1
	Discourse reference resolution in detail	May 2023	10	1
Module 6: Applications of NLP	Rule-based machine translation	Dec 2022	5	1
	Statistical approach to MT	Dec 2023	5	1
	Working of MT system	May 2024	10	1
	Text summarization	Dec 2022	10	1
	Information retrieval system	May 2024	10	1
	Text processing for IR	Dec 2022	5	1
	Question Answering system	Dec 2023	10	1
	IR vs IE system	May 2023	10	1